

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Powerful and secure SoCs for digital health applications

Ambiq has launched Apollo4 Lite and Blue Lite system on chips (SoCs), providing feature-rich capability, optimised memory, and powerful graphics performance. With secureSPOT technology, these SoCs deliver robust security



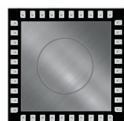
in a lightweight solution. The SoCs are suitable for digital health applications like stethoscopes and glucose and blood pressure monitoring. They provide a secure Bluetooth Low Energy (LE) connection to devices and the cloud. The chips have an ultra-low-power Cortex-M4 core operating up to 192MHz, effectively reducing overall system power consumption and extending device battery life. Both SoCs feature an audio subsystem, graphics processing unit (GPU), and abundant magnetoresistive random access memory (MRAM) and static random access memory (SRAM). The chips are designed for advanced security and powerful graphics in small form factors, which is crucial for device adoption.

Ambiq
<https://ambiq.com>

Award-winning edge AI processor boosts battery devices performance

NDP200 Neural Decision Processor (NDP) from Syntiant Corp. has been

awarded 'Best Product of the Year' by the tinyML Foundation. The processor architecture is designed for always-on-edge AI vision applications and allows full control of multiple independent

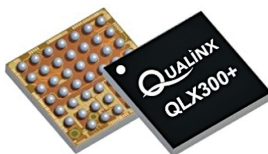


networks. It can operate concurrently at less than 2MW power with near-zero latency. This makes it suitable for tasks like person detection, object classification, motion tracking, and occupancy monitoring. The NDP200 offers precise neural processing for battery-powered consumer and commercial devices such as IP and security cameras, video doorbells, and teleconferencing equipment, among other applications. Utilising the Syntiant Core 2 programmable deep learning architecture, the NDP performs vision processing with highly accurate inference at under 1MW. Compared to its predecessor, the NDP200 delivers 25 times the tensor throughput, significantly boosting ML performance.

Syntiant Corp.
<https://www.syntiant.com>

World's tiniest and most efficient GNSS sensor

The QLX300+ launched by Qualinx stands as the tiniest and most energy-



efficient GNSS sensor globally, using up to one-tenth the energy of other existing devices. The sensor can simultaneously receive signals from navigation satellites like GPS, Galileo, GLONASS, and Beidou. This energy-efficient product extends battery life using zinc-air or single-cell batteries. Its Baseband Engine with 100+ reconfigu-

rable channels achieves high sensitivity and fast time-to-first fix (TTFF). The integrated low-noise amplifier (LNA) matching network eliminates the requirement for external components, leading to the lowest possible reduction in the bill of materials (BoM) cost. The product arrives in an industry-standard wafer-level chip-scale package, offering a single-chip solution in the smallest form factor. The sensor is an independent GNSS solution catering to diverse applications like wearables and asset/fleet/livestock tracking. Its compact size, ultra-low power consumption, and economical operation suit various needs, making it a prime option for connectivity and tracking across applications like smart-watches, healthcare monitoring, etc.

Qualinx
<https://qualinx.io>

Revolutionary AI data centre memory advancements

Micron Technology, Inc. has launched the industry's first 8-high 24GB HBM3

Gen2 memory, featuring bandwidth exceeding 1.2TB/s and pin speed surpassing 9.2GB/s. This marks a notable

enhancement of up to 50% compared to presently available solutions. With a 2.5-fold increase in performance per watt compared to earlier iterations, the HBM3 sets new records for critical artificial intelligence (AI) data centre measures such as performance, capacity, and power efficiency. These improvements reduce training durations for substantial language models like GPT-4 and onward, optimise AI inference infrastructure utilisation, and establish an

